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Reimagining Learning as a 3-Dimensional Vector: Integrating Behaviorism, Cognitivism, and Social Constructivism

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Abstract: This conceptual paper proposes the Three-Dimensional Learning Theory (3DLT) model as a new way of integrating three major perspectives on learning. While behaviorism, cognitivism, and social constructivism are often treated as separate paradigms in research, classroom practices reveal that learning naturally unfolds across all three. Drawing inspiration from physics and ongoing work in learning sciences, 3DLT conceptualizes learning as a vector whose behavioral, cognitive, and sociocultural components interact dynamically. The paper first revisits the historical fragmentation among learning theories, then introduces 3DLT as an integrative model, and finally demonstrates its presence in four common teaching methods: lecture-based, technology-based, group-based, and problem-solving learning. Through this analysis, the paper argues that effective teaching involves movement through interconnected dimensions rather than adherence to a single theoretical stance. Implications are offered for teaching, teacher education, and research design, positioning 3DLT as a framework for a more balanced and multidimensional understanding of how learning occurs.

The starting point: Beyond the silos of learning theories

Learning theories have long guided educational research and practice, yet they often operate in isolation. In most education programs, behaviorism, cognitive constructivism, and sociocultural theory are taught as distinct paradigms rather than interacting lenses on the single phenomenon: *learning*. This separation limits how we understand learning and how we design classrooms that reflect its complexity. Behaviorism emphasizes the *observable* (Pavlov, 1927; Skinner, 1953; Thorndike, 1932), cognitive constructivism stresses *mental processes* (Atkinson & Shiffrin, 1968; Chi & Wylie, 2014; Piaget, 1970), and sociocultural theory conceptualizes learning in *participation and belonging* (Lave & Wenger, 1991; Nasir & Hand, 2006; Vygotsky, 1978). Each contributes powerfully, but when viewed separately they often reduce learning to one dimension.

Sfard (1998) warned of the dangers of choosing just one metaphor for learning, a caution that continues to be ignored. In today's complex classrooms, where students use technology, collaborate, and solve real-world problems, these theories inevitably intertwine. Within any single degree, course, or classroom session, learners are expected to demonstrate competence through the intertwined logics of these theories. From a behaviorist perspective, students are evaluated through *observable* and *measurable* performances such as assignments and exams that reinforce desired outcomes (Ertmer & Newby, 1993; Skinner, 1953). From a cognitive view, learning involves encountering new information, experiencing cognitive dissonance, and reorganizing *mental schemas* through processes of assimilation and accommodation (Chi & Wylie, 2014; Piaget, 1970). Simultaneously, a sociocultural lens reminds us that as learners collaborate, engage in discourse, and participate in community practices, they develop new *identities* and a sense of *belonging* (Lave & Wenger, 1991; Vygotsky, 1978). Thus, the lived experience of learners already operates as a composite process—behavioral, cognitive, and social—woven together in their educational encounters.

In this conceptual paper, I argue that it is time to move beyond linear or siloed understandings, drawing inspiration from physics and other recent work happening in the field (NASEM, 2018; Nasir et al., 2020). I advocate for conceptualizing learning as a three-dimensional vector. The Three-Dimensional Learning Theory (3DLT) sees behavior, cognition, and social construction as orthogonal but interdependent axes of growth. Learning is not an isolated shift on one specific dimension but a dynamic movement through a multidimensional space where action, understanding, and participation interact. This idea draws inspiration from integrative thinkers who recognized the mutual shaping of body, mind, and environment (Cobb, 1994; Sfard, 1998). In proposing 3DLT, my goal is to provide a framework that captures the full richness of how students learn in contemporary classrooms and how educators and researchers can better operationalize trajectories of learning and growth.

In what follows, I briefly revisit the roots of behaviorism, cognitivism, and social constructivism to show how each offers a partial but important view of the complex phenomenon of learning. Although many other perspectives inform educational theory, this paper focuses on these three for simplicity. I then propose the 3DLT and use examples from classroom practices to illustrate how different teaching methods already blend these perspectives. I conclude by discussing how this framework can enrich teaching and research.

The problem: Theoretical fragmentation

The persistence of fragmentation among learning theories is both historical and epistemological. Early educational psychology was dominated by behaviorism, which defined learning as a measurable change in behavior (Bandura, 1977; Pavlov, 1927; Skinner, 1953; Thorndike, 1932). Pavlov's classical conditioning, Thorndike's "law of effect," Skinner's operant conditioning, and Bandura's observational learning grounded an entire generation of teaching practices such as drill, reinforcement, and programmed instruction (Ertmer & Newby, 1993).

Yet as Lagemann (2000) noted, this scientific pursuit of control often neglected mental life. The cognitive revolution re-centered the learner's mind, shifting attention toward how information is processed, organized, and transformed (Miller, 1956; Piaget, 1970). Researchers such as Chi and Wylie (2014) extended this view through the Interactive–Constructive–Active–Passive (ICAP) framework, which describes how deeper levels of engagement correspond to more meaningful learning. Despite this progress, cognitive constructivism was still criticized for portraying learners as isolated thinkers, detached from the social and cultural contexts that shape understanding and belonging.

Sociocultural theorists responded by embedding the cognitivist learning model within human relationships. Vygotsky's (1978) *Zone of Proximal Development* framed cognition as emerging first between people (interpsychological) and then within the individual (intrapsychological). Lave and Wenger (1991) extended this to communities of practice, while Cobb (1994) argued that learning is both social and individual, not either-or.

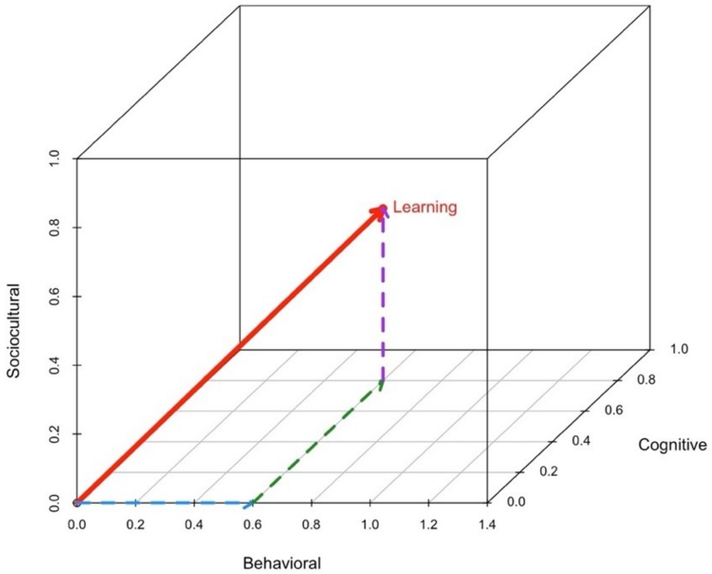
Sfard's (1998) critique of metaphoric dualism—the acquisition metaphor versus the participation metaphor—captures the root problem. When researchers align exclusively with one lens, they risk flattening learning into one plane of reality. Behaviorism examines the external plane of performance, cognitive constructivism the internal plane of reasoning, and sociocultural theory the interpersonal plane of participation. Each observes a projection of the learning phenomenon rather than its totality. Although each theory has its affordances and constraints, when one is preferred in isolation, the result is one-dimensional thinking in instructional design, teaching, or research.

To transcend this, we may treat learning theories not as rival explanations but as an integrated coordinate system. Just as the motion of an object in three-dimensional space cannot be fully described by a single axis, the complex phenomenon of learning cannot be captured through a preferred axis of behavior, cognition, or social interaction alone. The fragmented landscape of theories reflects our analytic convenience, not the learner's lived experiences. Recognizing this limitation sets the stage for an integrated model—one that allows educators and researchers to navigate and examine, rather than choose between, the multiple aspects that characterize learning.

The proposal: Learning as a three-dimensional vector

The Three-Dimensional Learning Theory (3DLT) conceptualizes learning as a vector with three perpendicular yet interdependent axes: behavioral, cognitive, and sociocultural (see Figure 1).

Figure 1
The Three-Dimensional Learning Vector



The behavioral axis represents observable actions shaped through reinforcement, feedback, and habit (Ertmer & Newby, 1993; Skinner, 1953; Thorndike, 1932). It captures fluency, accuracy, and skill automatization—processes still vital for foundational mastery (Goldman & Pellegrino, 2015). The cognitive axis reflects internal organization of knowledge, the construction of mental models, and metacognitive control (Chi & Wylie, 2014; Piaget, 1970; Schoenfeld, 1985). It denotes depth of understanding and the learner's ability to integrate new information with prior *schemas*. The sociocultural axis embodies interaction, participation, and identity formation within communities of practice (Cobb, 1994; Lave & Wenger, 1991; Vygotsky, 1978).

In this model, a learner's state of learning can be represented by a position vector $L\ (x, y, z)$, where x, y, and z correspond to behavioral performance, cognitive development, and social participation. The magnitude of L reflects the overall strength of learning, while its direction reflects how learning is distributed across these three components. Teaching moves learners through this space by differentially activating the axes: reinforcement may build fluency, reflection may deepen understanding, and collaboration may strengthen participation, recognition, and belonging. Learning growth, then, is not the triumph of one theory over another but their combined resultant.

The 3DLT model can be useful for both teaching and research. Consider a teacher designing a lesson on linear equations. One goal may be procedural fluency in solving equations, another a deeper grasp of what slope and intercept represent, and a third the creation of opportunities for students to explain, collaborate, and begin to see themselves as "math people." Researchers, likewise, can use the 3DLT model to trace trajectories of learning across contexts by examining evidence of growth along the three dimensions. However, how each dimension is operationalized will depend on the research question and context.

The evidence: Teaching methods as three-dimensional interplay

To ground this theoretical proposal in classroom reality, I turn to four common teaching practices that reveal how learning naturally unfolds across behavioral, cognitive, and social dimensions. These four approaches—lecture-based, technology-based, group-based, and problem-solving—illustrate that the 3DLT is not a distant abstraction but already present in everyday instruction. By re-examining each method through this three-dimensional lens, I show how teachers, often implicitly, navigate multiple theories at once and how these blended practices exemplify learning as movement through an integrated space rather than along isolated lines.

Before turning to the analysis, it is useful to clarify what these four teaching/learning methods entail. Lecture-based learning refers to direct instruction where teachers primarily present information to students, sometimes supplemented by brief interactions or demonstrations (Bligh, 2000). Technology-based learning (TBL) encompasses the use of digital tools such as simulations, online platforms, or multimedia resources to enhance access and engagement (Chinn & Iordanou, 2023; Koller et al., 2006). Group-based learning involves students working collaboratively toward shared goals, emphasizing dialogue, accountability, and peer explanation (Johnson & Johnson, 1999; Webb, 2009). Problem-solving centers on engaging learners with complex, often open-ended tasks that promote reasoning, strategy use, and creativity (Jonassen, 2000; Schoenfeld, 1985). In what follows, I re-examine each of these methods through the lens of the Three-Dimensional Learning Theory (3DLT) to show how behaviorism, cognitivism, and social constructivism operate together in everyday classroom practice.

Lecture-based learning

Lecture-based learning, while often labeled behaviorist, also draws on cognitive and social dimensions. A teacher modeling equations and giving recall quizzes clearly uses reinforcement and feedback (Ertmer & Newby, 1993; Skinner, 1953). Yet when that teacher employs analogies, visual organizers, or prediction questions, they activate cognitive schemas (Piaget, 1970). Adding think-pair-share discussions transforms the lecture into a social activity aligned with Vygotsky's (1978) *Zone of Proximal Development* and Cobb's (1994) participation view. Deslauriers, Schelew, and Wieman (2011) empirically showed that active, socially engaging lectures outperform traditional ones—evidence that integrating all three dimensions enhances learning.

Technology-based learning

Likewise, technology-based learning (TBL) operates in this 3D space. Behaviorist design appears in drill-and-practice software offering immediate feedback and rewards (Ertmer & Newby, 1993; Koller et al., 2006; Skinner, 1953). Cognitivists emphasize multimedia principles and mental load management (Mayer, 2004), while sociocultural theorists highlight online collaboration and shared knowledge construction (Sfard, 1998; Vygotsky, 1978). Clark (1983) argued that media are merely delivery tools, yet contemporary evidence (Chinn & Iordanou, 2023) shows that technology's value lies in how it shapes interaction and reasoning. Thus, effective TBL situates learners along all three axes—practice, cognition, and participation.

Group-based learning

Group-based learning foregrounds the sociocultural axis but remains multidimensional. Cooperative learning succeeds when interdependence is reinforced behaviorally (Ertmer & Newby, 1993; Johnson & Johnson, 1999), when peers elaborate ideas cognitively (Chi & Wylie, 2014; Piaget, 1970), and when communities of practice form socially (Bielaczyc & Collins, 1999; Vygotsky, 1978). Research confirms that explanation and argumentation drive conceptual change during collaborative work (Webb, 2009). 3DLT interprets these not as parallel benefits but as interactive components of one learning vector.

Problem solving

Finally, problem-solving epitomizes three-dimensional learning. Behavioral elements appear in guided practice and reinforcement of correct strategies. Cognitive processes dominate through metacognition, productive failure, and transfer (Kapur, 2016; Schoenfeld, 1985). Sociocultural elements emerge in collaborative inquiry and authentic contexts (Cobb, 1994; Sfard, 1998; Vygotsky, 1978). In science or math classrooms, students negotiating a shared experiment embodies all three dimensions: they apply procedures, reason conceptually, and engage socially. The strength of learning arises from the combined magnitude of these vector components, not from allegiance to a single theory. Table 1 below summarizes how each teaching method draws on behaviorism, cognitivism, and social constructivism, showing that effective instruction rarely aligns with a single theory.

Table 1
Teaching methods and their connection to learning theories.

	Behaviorism	Cognitivism	Social Constructivism
Lecture-based learning	Reinforcement	Organization/Schema	Dialogue
Tech-based learning	Feedback	Modeling/Visual	Virtual Collaboration
Group-based learning	Accountability	Explanation/Elaboration	Community
Problem-solving	Practice	Metacognition/Strategy	Negotiation

The implications: Towards an integrated idea of learning

Viewing learning as a three-dimensional vector carries theoretical, methodological, and pedagogical implications. Theoretically, it transforms educational discourse from comparative to compositional. Rather than debating whether behaviorism or constructivism is correct, 3DLT suggests that each capture an important aspect of the same phenomenon. This mirrors Goldman and Pellegrino's (2015) call for integrative perspectives linking curriculum, instruction, and assessment. Methodologically, 3DLT invites mixed analyses that track behavioral data (performance), cognitive indicators (reasoning patterns), and sociocultural evidence (interaction and discourse). Such triangulation echoes the blended approaches in contemporary learning sciences (Chinn & Iordanou, 2023).

Pedagogically, 3DLT empowers teachers to design for balance. A math teacher emphasizing drill (x-axis) can intentionally add reflection prompts (y-axis) and peer discussion (z-axis) to move students diagonally through the learning space. A science educator using group inquiry might incorporate structured reflection and reinforcement to stabilize all axes. The new shifts do not abandon any of the three theories but orchestrate their interdependence. As Bligh (2000) and Deslauriers et al. (2011) showed, even traditional formats become transformative when re-engineered through multidimensional engagement.

In research, scholars could model trajectories of learning vectors across time, mapping how interventions alter the direction and magnitude of $L\ (x,\ y,\ z)$. This may connect micro-level behavior with macro-level participation, aligning with Vygotsky's (1978) developmental vision and Lave and Wenger's (1991) situated learning. Ultimately, 3DLT turns siloed learning theories into a unified spatial theory of transformation.

In conclusion, the 3DLT model recognizes what educators intuitively enact: learning is simultaneously behavioral, cognitive, and social. While the learning sciences have increasingly called for integration, this framework provides a formalized, geometric language to map these ideas. Rather than forcing a choice between paradigms, the field's task is to chart the full vector of human growth. As a brief conceptual paper, this work leaves several aspects of the model for further development. Most notably, learning is ultimately an n-dimensional phenomenon; future iterations of this coordinate system could expand to include other important axes to capture the full richness of a learner's development. Ultimately, I hope 3DLT proves useful as a learning heuristic for researchers, a design heuristic for instructors, and a communicative bridge for teacher educators seeking to operationalize the multidimensional reality of meaningful education.

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